Endurance and self-noise of the LC-Mare recorder 1B843C2E

I. INTRODUCTION

A passive acoustic recorder is characterized by two properties that the deployment cannot correct afterwards: how long it records without loss on one battery, and how much noise it adds to the signal. This report presents both for the LC-Mare unit 1B843C2E. An endurance test of 35.6 days established the recording time, the data volume and the stability of the self-noise over the discharge of the battery. A measurement with resistors in place of the hydrophone, following the method of the PAM book (ch. 11, Preamplifier noise and gain), established the gain and the input noise of the front end, and with them the calibration of the recordings in equivalent sound pressure.

The numbers and figures are produced by three scripts in `tests/`: `report_1B843C2E.py` for the endurance test, `report_resistor_1B843C2E.py` for the resistor measurement, and `characterization_1B843C2E.py` for the combined spectra of Fig. 4.

II. MATERIALS AND METHODS

A. Recorder

The recorder consists of an SQ17-00 hydrophone, a differential preamplifier with a pair of AD8656 operational amplifiers and a nominal gain of 21, a TLV320ADC6140 sigma-delta converter, and an Adafruit Feather RP2040 Adalogger running firmware 3.0.2 (`Adalogger_V3a`). For both measurements, the unit recorded one channel at 96 kHz with a programmable gain of 20 dB, in compressed mode (PROC_MODE 1, SHIFT 12), and wrote one file per minute to a 512 GB exFAT microSD card with 256 KiB clusters. The converter delivers 24-bit samples left-aligned in 32-bit words, which the firmware shifts right by 12 bits, so that full scale is 2^{19} counts. Times are board times, as in the file names.

B. Endurance test

The endurance test started at 2026-09-17 11:06:03, after a battery change, and ended at 2026-10-23 00:29:00, the end of the last file holding data, an elapsed time of 3 072 177 s, or 35.56 days. The recorder was placed in air inside an aluminum tube, which served as a Faraday cage. The recordings therefore contain no underwater sound. A quiet room of 30 dB(A) re 20 µPa has a spectral level of about 16 dB re 1 µPa²/Hz near 10 kHz, some 20 dB below the floor reported in Sec. III.C before any attenuation by the tube, so that the recordings show the self-noise of the system together with any interference generated inside the recorder. Three files were analyzed in detail (Table I).

Table I. Files of the endurance test analyzed in detail.

	File	Header voltage	Selection
A	`1B843C2E_20260917_235900.bin`	4.35 V	last file of the first day
B	`1B843C2E_20261013_185900.bin`	3.39 V	last file of the first hour with a median of 3.40 V, before the reading saturates near 3.3 V
C	`1B843C2E_20261022_235900.bin`	3.36 V	last file of the day before the last

C. Resistor measurement

On 25 October 2026, the hydrophone was replaced by a short circuit and by resistors of 978 Ω, 49.5 kΩ, 98.9 kΩ and 299 kΩ across the differential input of the preamplifier, and each configuration was recorded for 13 to 20 complete one-minute files. The value is stored in the ISRC chunk of each file. Two files were excluded: the first full file of the 98.9 kΩ session, which reports one missed acquisition and carries a transient, and its last file, which clips when the resistor was removed.

The power spectral density of the recorded noise is modeled after the PAM book (Eq. 11-lcm-noise) as

P(f, R) = G²(f) {e_n²(f) + 4 k_B T Re Z(f, R)}, Z = R ∥ R_in ∥ 1/(j2πfC), (1)

where P is in counts²/Hz, G is the voltage gain of the whole system from the preamplifier input to the sample, in counts/V, e_n is the input noise voltage density, k_B is the Boltzmann constant, T = 293 K, R is the test resistor, R_in = 4 MΩ is the differential input resistance formed by the two bias resistors of 2 MΩ, and C is the capacitance across the input. The input noise current of the CMOS amplifiers is neglected. For R ≪ R_in and 2πfRC ≪ 1, Eq. (1) reduces to the form of the book. For a given C, Eq. (1) is linear in G² and G²e_n², which were estimated for every frequency bin by weighted least squares over the five resistors. C was chosen to minimize the residual between 2 and 15 kHz, and the values quoted below are medians over this band, unless a frequency is given.

D. Spectral analysis and calibration

All spectra are Welch estimates with a Hann window of 4096 points, i.e. a resolution of 23.4 Hz, and 50 % overlap. The resistor spectra are averaged over all files of a configuration.

The samples are converted to equivalent sound pressure by

L_p = 20 log10|I| − G_sys − M_dB, (2)

where I is the sample in counts, G_sys = 158.2 dB re 1 count/V is the system gain measured with the resistors (Sec. III.B), and M_dB = −201 dB re 1 V/µPa is the low-frequency receiving sensitivity of the SQ17-00 taken from its response curve, as in the front-end model of the PAM book. One count then corresponds to 42.8 dB re 1 µPa, i.e. 0.138 mPa, or 12.3 nV at the preamplifier input, and full scale to about 157.2 dB re 1 µPa peak. A spectral level in dB re 1 count²/Hz is converted to dB re 1 µPa²/Hz by adding 42.8 dB. With the nominal sensitivity of −202.5 dB re 1 V/µPa of the data sheet, all levels would be 1.5 dB higher.

Eq. (2) assumes a flat sensitivity, which according to the model of the book holds within 1 dB from 137 Hz to 40 kHz. Above 40 kHz the resonances of the hydrophone shape the sensitivity. As the endurance test was made in air, the levels in µPa are the equivalent noise pressure that the system would add in water, and not a sound pressure in air. For the resistor recordings, the levels in µPa express the input voltage in the same units, so that both kinds of spectra may be compared directly (Sec. III.C).

III. RESULTS AND DISCUSSION

A. Endurance

The unit recorded without interruption for 35.56 days on one battery (Table II). Of the 51 204 files written, 51 203 hold data, which is exactly the number expected at one file per minute,

and the only empty file was opened when the battery was exhausted. The records of all files add up to 3 072 190 s of audio, which agrees with the elapsed time to within 13 s, or 4 ppm. Some files hold 5607 instead of 5628 records, i.e. 0.22 s less, although their headers report no missed acquisition. This is an effect of the file rotation and not a loss of data, as the agreement of the total shows. The battery voltage read from the file headers fell from about 4.4 V at the start to 3.4 V on 13 October, after which the reading saturated near 3.35 V for the last nine days (Fig. 1). The voltage at the end of the test is therefore not resolved by the header reading.

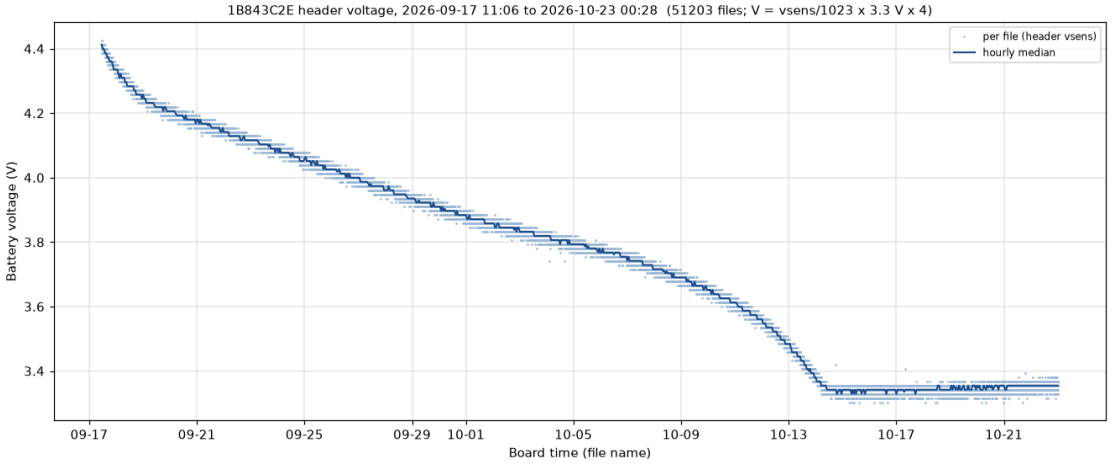


Fig. 1. Battery voltage of the endurance test against board time, read from the header of every file as V = vsens/1023 × 3.3 V × 4. Dots: individual files; line: hourly median. The quantization steps of about 13 mV are those of the 10-bit reading.

Table II. Data volume of the endurance test.

Quantity	Value
Files	51 204, of which 51 203 hold data
Data written	442.42 GB
Allocated on disk (256 KiB clusters)	442.97 GB, i.e. 0.12 % overhead
Mean bit rate	1152 kbit/s, i.e. 12.0 bit per sample
Bit rate of raw mode (int32)	3072 kbit/s
Compression ratio	2.67

The mean bit rate was 1151 to 1152 kbit/s on every day, and it is set by the record format rather than by the signal. The compressor used 10 or 11 bit per sample, i.e. 10.4 to 10.5 bit per sample including the record header. However, every record is padded to a multiple of 128 words, so that a record of 1024 samples with 325 to 357 words of payload always occupies 384 words. The padding therefore costs 13 % of the storage, and without it the mean bit rate would be about 1004 kbit/s.

B. Gain and input noise

The noise spectra of the five resistors are stationary to within the precision of the estimate: in the band from 2 to 15 kHz, the levels of the individual files of a configuration differ by at most 0.06 dB. Eq. (1) fits them from about 100 Hz to 15 kHz with an rms relative residual of 1.1 %, and the median residual of each resistor is at most 0.03 dB (Fig. 2).

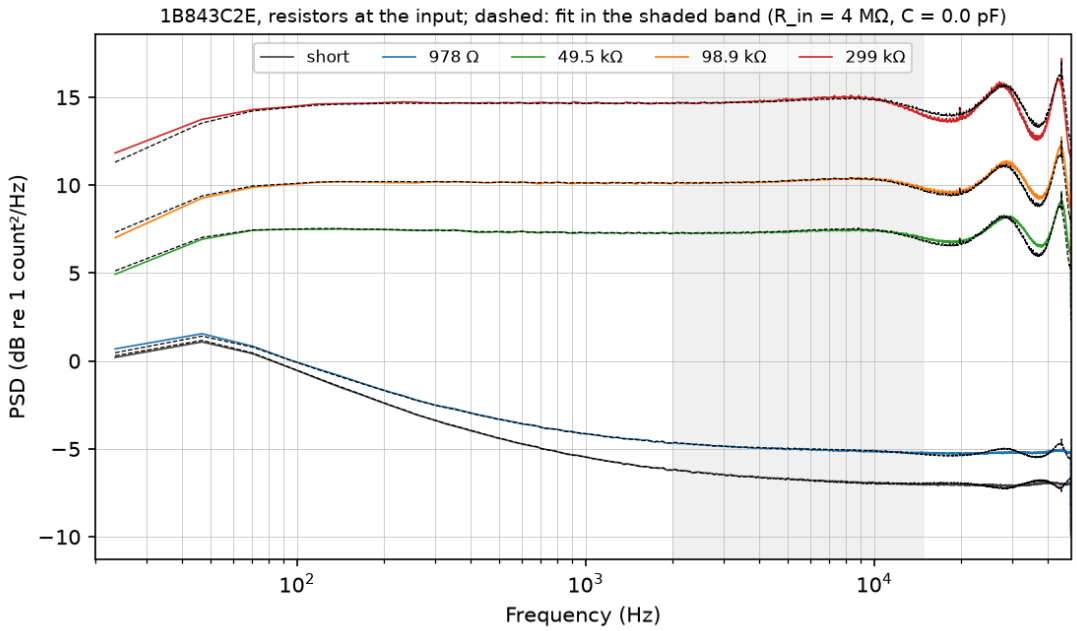


Fig. 2. Power spectral density of the recorded noise in dB re 1 count²/Hz for the short circuit and the four resistors across the preamplifier input, each averaged over 13 to 20 one-minute files. Dashed: fit of Eq. (1) with R_in = 4 MΩ and C = 0 in the shaded band from 2 to 15 kHz, extended over the whole frequency range.

The system gain is G_sys = 158.2 dB re 1 count/V, flat within ±0.15 dB from 200 Hz to 15 kHz. It falls by 0.5 dB at 100 Hz and by about 4 dB in the lowest bin at 23 Hz, which is consistent with the digital high-pass filter of the converter at 24 Hz (Fig. 3, upper panel). The input noise is e_n = 5.6 nV/√Hz between 2 and 15 kHz and 5.45 nV/√Hz at 10 kHz, which agrees with the 5.4 nV/√Hz measured on the LC-Mare preamplifier in the PAM book and the 5.5 nV/√Hz of its model. Toward low frequencies e_n rises to 6.65 nV/√Hz at 1 kHz, 9.4 nV/√Hz at 200 Hz and 12.5 nV/√Hz at 100 Hz, which corresponds to a flicker-noise corner near 0.5 kHz (Fig. 3, lower panel).

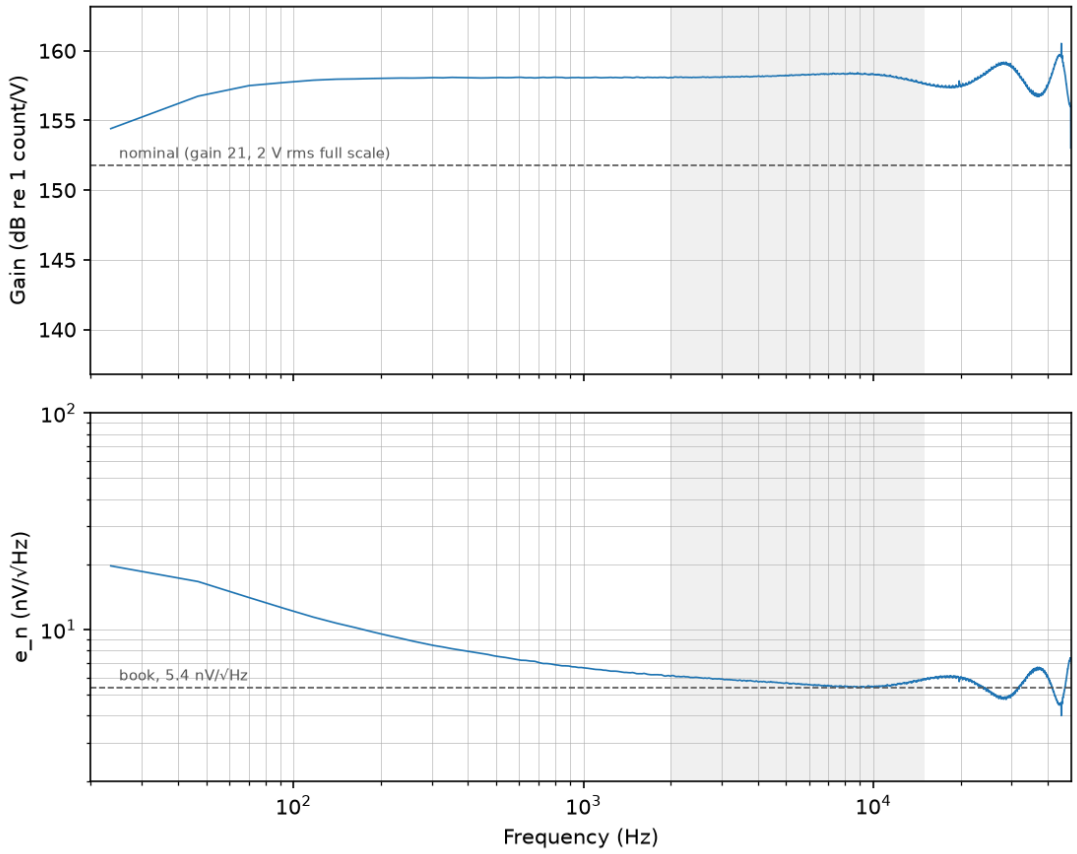


Fig. 3. System gain G in dB re 1 count/V (upper panel) and input noise voltage density e_n in nV/√Hz (lower panel), estimated per frequency bin from Eq. (1). Dashed: nominal gain for a preamplifier gain of 21, 20 dB of programmable gain and 2 V rms differential full scale (upper), and the input noise of 5.4 nV/√Hz measured in the PAM book (lower). Shaded: band from 2 to 15 kHz used for the fit. Above 15 kHz, the ripple of both curves is an artifact of the model (Sec. III.B) and not a ripple of the gain.

The result is insensitive to the details of the model. The estimator of the book (Eq. 11-lcm-en), which ignores R_in and C, gives 158.23 dB and 5.60 nV/√Hz, and a fit of the three smallest resistors only, 0 Ω to 49.5 kΩ, for which neither R_in nor C matters, gives 158.18 dB and 5.59 nV/√Hz. R_in and C cannot be separated, as both lower the noise of the largest resistor: with R_in = 4 MΩ the best fit is found for C = 0, and without bias resistors for C = 13.7 pF. Across these cases, and for R_in = 2 MΩ, the gain varies by 0.23 dB and e_n by 0.14 nV/√Hz. An error of 5 K in the assumed temperature changes the gain by 0.04 dB.

Above 15 kHz, the spectra of the kΩ resistors show humps near 28 and 44 kHz of about 1.5 dB, which the short circuit and the 978 Ω resistor do not show (Fig. 2). As the humps scale with the resistor noise rather than with the input noise, they cannot arise in the converter or its digital filters, which would shape both alike. They are therefore likely a response of the preamplifier to a high source impedance, for example of the amplifier input capacitance together with the feedback capacitors of 15 pF, and Eq. (1) does not separate gain and input noise there. The source impedance of the hydrophone, 3.5 kΩ at 28 kHz, is an order of magnitude below that

of the smallest resistor that shows the humps, and one may expect the effect on hydrophone recordings to be small, but the present data cannot confirm this.

The measured gain may be compared with the nominal gain that follows from the component values, as in the system calibration of the LC-Mare front end in the PAM book (ch. 11),

G_nom = 20 log10 I_fs − 20 log10 V_fsc + 20 log10 G_0 + G_PGA = 114.4 − 9.0 + 26.4 + 20 = 151.8 dB re 1 count/V, (3)

where I_fs = 2^{19} counts is the full-scale value of the samples after the shift by 12 bits, V_fsc = $2\sqrt{2}$ V = 2.83 V the peak differential full scale of the TLV320ADC6140 at 0 dB programmable gain, i.e. 2 V rms, G_0 = 1 + (R_2 + R_3)/R_1 = 21 the differential gain of the preamplifier, and G_PGA = 20 dB the programmable gain read from the file header. The full scale of 2 V rms is the one that the register settings of the firmware select (line input CH1_INSRC = 00, ADC_FSCALE = 0, i.e. VREF = 2.75 V). Nominally, one count therefore corresponds to 10^(−151.8/20) V = 25.7 nV at the preamplifier input, or 49.2 dB re 1 µPa with the sensitivity of Eq. (2).

The measured gain of 158.2 dB re 1 count/V is 6.4 dB, or a factor of 10^(6.4/20) = 2.09, above the nominal gain of Eq. (3), and one count corresponds to 12.3 nV instead of 25.7 nV. As the endurance recordings clip at ±2^{19} counts, a shift of the data by one bit can be excluded, and the factor must arise in the analog part. Two explanations are possible, and Eq. (3) shows by how much each term would have to change. If the preamplifier gain is the nominal 21, the measured gain requires V_fsc = 2.83 V / 2.09 = 1.35 V peak, i.e. 0.96 V rms. This is close to the full scale of 1 V rms that the converter has with VREF = 1.375 V or in single-ended mode, and with exactly 1 V rms the preamplifier gain would be 21.9, 4 % above nominal. If, on the other hand, the full scale is the nominal 2 V rms, the preamplifier gain must be 21 × 2.09 = 43.9. With R_2 + R_3 = 20 kΩ, this requires R_1 = 20 kΩ / 42.9 = 466 Ω instead of 1 kΩ, which a resistor of 470 Ω would explain. A check of R_1 on the board, or a sine of known amplitude applied directly to the converter input, would decide between the two. For the calibration, however, only the product matters, and Eq. (2) uses the measured one.

C. Self-noise

Fig. 4 overlays the spectra of the endurance test and of the resistor measurement in equivalent noise pressure. Between 10 and 40 kHz, the short circuit gives 35.8 dB re 1 µPa²/Hz and the hydrophone 36.0 dB (file A). The self-noise of the system with the hydrophone therefore lies only 0.2 dB above that of the amplifier alone, and it agrees with the 36 dB that the model of the book predicts for this configuration. The converter contributes little: even at 0 dB programmable gain its noise of 40 nV/√Hz amounts to 1.9 nV/√Hz at the preamplifier input, and at 20 dB less still.

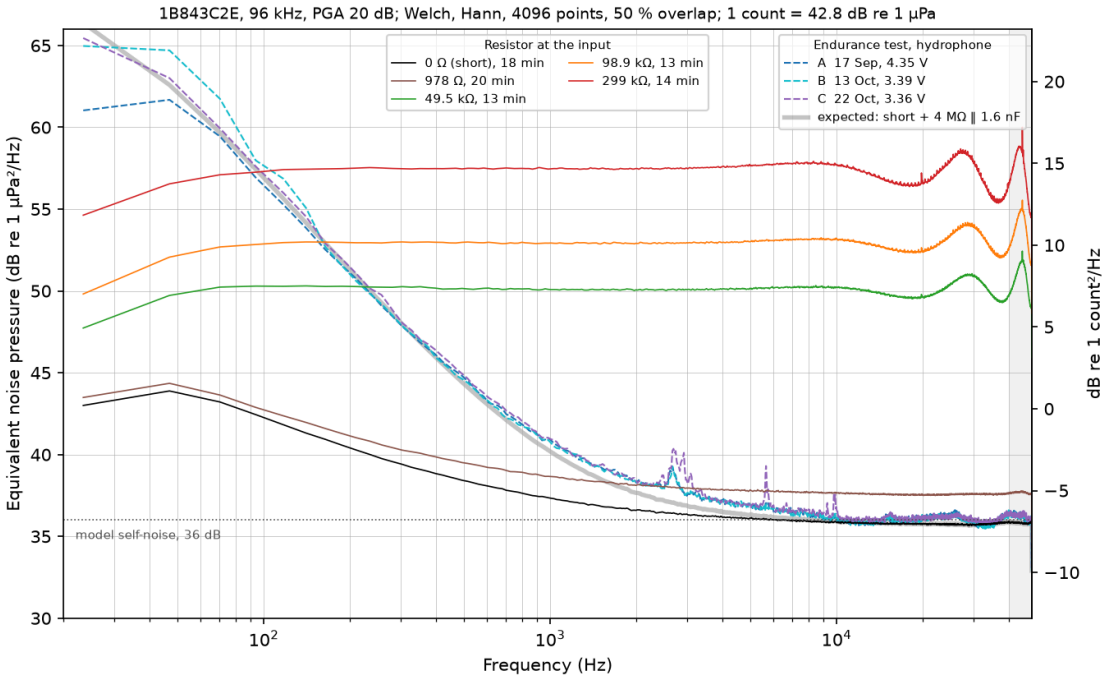


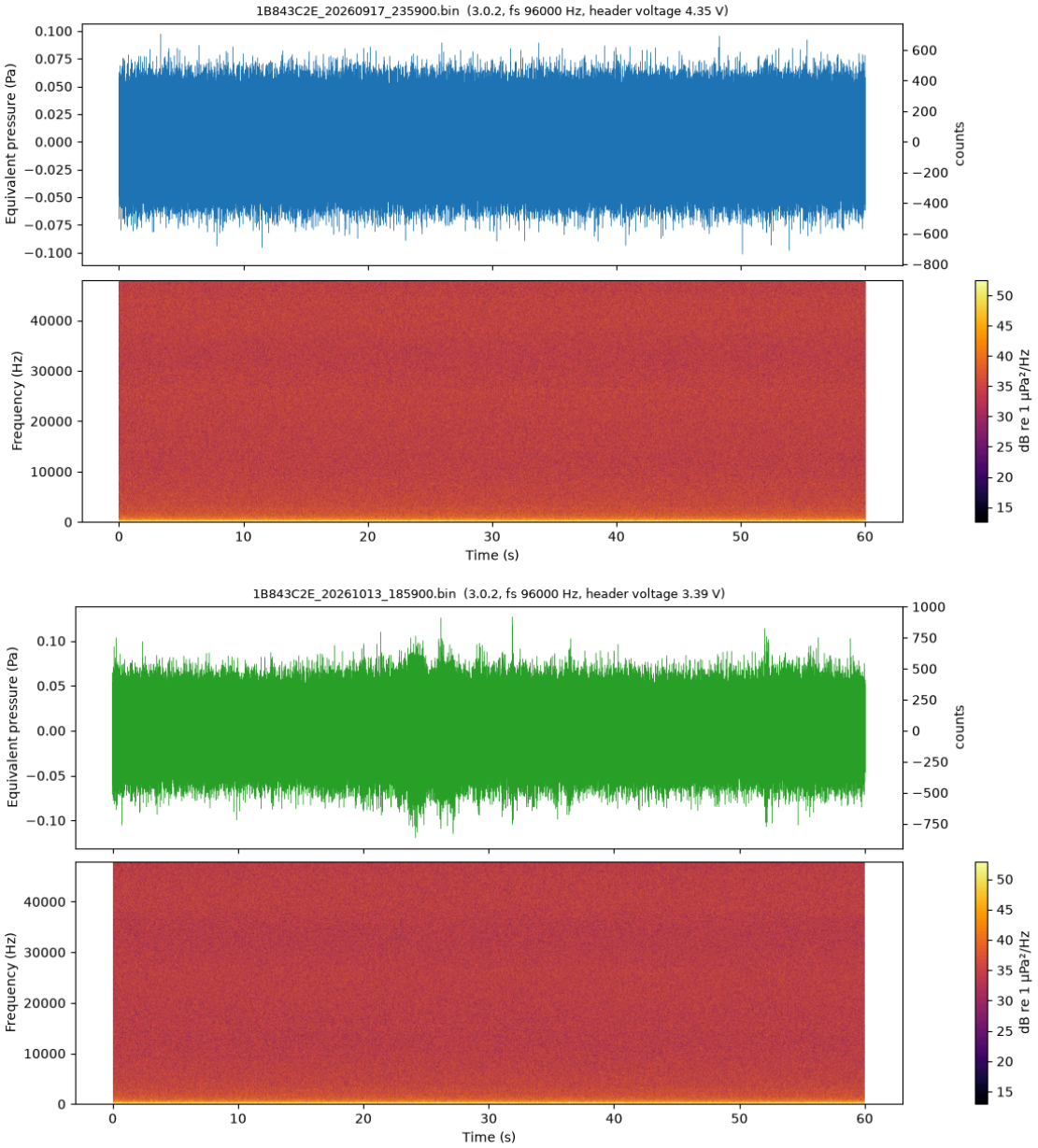
Fig. 4. Power spectral density in equivalent noise pressure, dB re 1 µPa²/Hz (left axis), and in dB re 1 count²/Hz (right axis), with one count = 42.8 dB re 1 µPa. Dashed: files A, B and C of the endurance test with the hydrophone connected (Table I). Solid: resistors of 0 Ω to 299 kΩ across the preamplifier input, averaged over 13 to 20 minutes each. Thick gray: spectrum expected with the hydrophone, the short-circuit spectrum plus the thermal noise of the input resistance R_in = 4 MΩ shunted by the hydrophone capacitance C_H = 1.6 nF. Dotted: self-noise of the model of the PAM book. Shaded: above 40 kHz, where the hydrophone resonances shape the sensitivity.

Below about 2 kHz the hydrophone spectra rise far above the short circuit, to 40.9 dB at 1 kHz and 51.1 dB re 1 µPa²/Hz at 200 Hz, against 37.3 and 40.4 dB. This excess is the thermal noise of the input resistance R_in, which the hydrophone capacitance C_H shunts only at high frequencies. Its real part, $R_{in}/\{1 + (2\pi f R_{in} C_H)^2\}$, is 61 kΩ at 200 Hz, which places the hydrophone spectrum between those of 49.5 and 98.9 kΩ, as Fig. 4 shows, and it falls to 25 Ω at 10 kHz. It should be emphasized that the expected spectrum in Fig. 4 contains no fitted parameter: it gives 51.3 dB at 200 Hz and 40.2 dB at 1 kHz, against 51.1 and 40.9 dB measured. The remaining excess of up to about 0.7 dB between 1 and 5 kHz may relate to the dielectric losses of the hydrophone, which the model omits. Consequently, the self-noise of the recorder is set by the preamplifier above about 3 kHz and by the bias resistors below. Larger bias resistors would lower the latter, at the cost of a higher sensitivity to leakage currents (PAM book, ch. 11).

D. Stability over the battery discharge

The self-noise did not change over the test. Between 5 and 45 kHz, the spectrum of file B lies a median 0.04 dB below that of file A (interquartile range −0.12 to 0.05 dB) and that of file C a median 0.07 dB above it (−0.04 to 0.18 dB); below 500 Hz the differences are 0.05 and 0.3 dB. All three files are dominated by stationary broadband noise, with standard deviations of 136, 154 and 171 counts, or 19, 21 and 24 mPa equivalent, and the spectrograms show no

distinct transients (Fig. 5). The larger standard deviations of files B and C do not contradict the matching spectra. Split into segments of 43 ms, the segment means scatter by 15, 17 and 65 counts in files A, B and C, and the 99th percentile of the segment RMS is 161, 243 and 258 counts. File B therefore already carries occasional loud segments, whereas only file C carries additional energy below the first Welch bin, and both contributions are averaged out of the Welch spectrum. The median segment RMS of files B and C lies only 0.7 and 0.5 dB above that of file A.



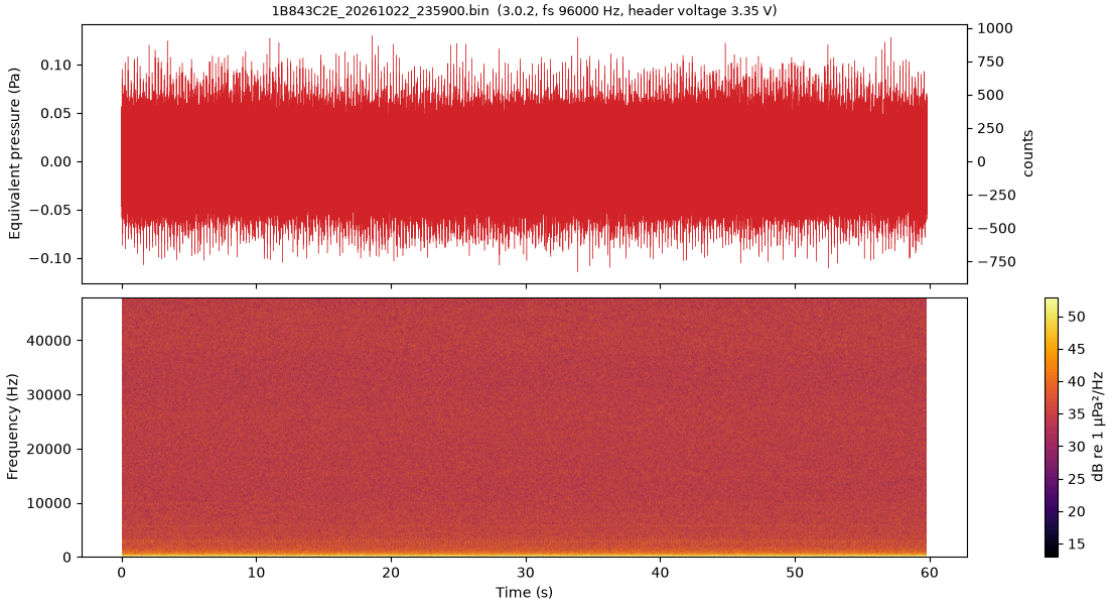


Fig. 5. Files A (a), B (b) and C (c) of the endurance test (Table I). Upper panels: time series in equivalent pressure, Pa (left axis), and in counts (right axis). Lower panels: spectrogram in dB re 1 µPa²/Hz, Hann window of 1024 points with 50 % overlap, with a color range of 40 dB below the 99.9th percentile.

All three hydrophone spectra show a broad hump of about 1 dB near 2.7 kHz, which none of the resistor spectra show (Fig. 4). Only in file C does it carry narrow tonal lines at 2695 and 2742 Hz, about 2 dB above the floor, together with a line at 5672 Hz and weaker lines near 2.9 and 9.8 kHz. As the recorder was screened by the aluminum tube, these lines cannot be electromagnetic pickup from outside. They are therefore likely generated inside the recorder, which supports the contention that they relate to the supply falling out of regulation at the end of the battery, but three files are not sufficient to establish this. The hump itself, present with the hydrophone only, may be sound or vibration reaching the hydrophone, which the present data cannot identify.

IV. CONCLUSIONS

The unit recorded without interruption for 35.6 days on one battery, writing 51 203 one-minute files and 442 GB at 96 kHz with no acquisition lost. In compressed mode it stores 12 bit per sample, a compression ratio of 2.67 against raw mode, whereby the padding of the records rather than the signal fixes the bit rate.

The system gain, measured with resistors at the preamplifier input, is 158.2 dB re 1 count/V, flat within ±0.15 dB from 200 Hz to 15 kHz, and with the hydrophone sensitivity of −201 dB re 1 V/µPa one count corresponds to 42.8 dB re 1 µPa. The gain is a factor of 2.09 above the nominal 151.8 dB re 1 count/V of Eq. (3), obtained from a preamplifier gain of 21, 20 dB of programmable gain and a converter full scale of 2 V rms at 2^{19} counts, which leaves open whether the preamplifier gain or the full scale of the converter deviates, but does not affect the calibration. The input noise of 5.6 nV/√Hz agrees with the PAM book. With the hydrophone, the self-noise is 36.0 dB re 1 µPa²/Hz between 10 and 40 kHz, as the model predicts, and it rises below about 3 kHz with the thermal noise of the bias resistors, which a model without

free parameters reproduces to within 0.7 dB. The self-noise stayed the same to within 0.1 dB from the first full day of the test to the last.

The separation of gain and input noise breaks down above 15 kHz, where the noise of large source resistors shows humps that the short circuit does not. A measurement with a capacitor equal to that of the hydrophone in place of the resistors would characterize the front end there under the source impedance it sees in use. A scan of all files between 13 and 22 October for the tonal lines of file C would date their onset relative to the battery voltage.